

Maharlika Store
A Centralized Outsourcing and Promotional Platform
for Local Retail Shops



A System Proposal Presented to the Faculty of the
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SYSTEM ANALYSIS AND DESIGN

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Part I

SYSTEM ANALYSIS REPORT

Introduction

The rapid growth of e-commerce and digital payment platforms has transformed the retail industry in the Philippines. However, many small to medium-sized retail shops in Davao City still rely on manual processes for inventory management, sales tracking, and customer collections. These manual methods often lead to stock discrepancies, delayed reporting, and difficulty in managing multiple payment schemes such as layaway, paluwagan, and installments.

Maharlika Store is a proposed centralized outsourcing and promotional platform designed to help local retail shops, particularly in Davao City, gain online visibility and streamline their internal operations. The platform functions similarly to the iTunes Store, where multiple shops can be listed, discovered, and promoted to potential customers. For shop owners, Maharlika Store provides an integrated Point-of-Sale (POS) and inventory management system that automates stock tracking, sales recording, and collection monitoring across various payment schemes including buy-and-sell, trade-in (upgrade/downgrade), paluwagan, layaway, credit card installment, GGives, Skyro, and investment services.

This Systems Analysis Report documents the current business operations of the primary partner shop, Maharlika Marexx Republic Davao (operating as Maharlika Gadgets), identifies underlying problems and opportunities, and proposes an automated solution through the Maharlika Store platform. The analysis was conducted through direct observation, client interviews with two owners (Jerold Yu and Mary Jane Yu) and two staff members, and examination of existing manual records using Excel spreadsheets. This report serves as the foundation for designing and developing the Maharlika Store platform as a capstone project.

The Organization

Maharlika Marexx Republic Davao is a sole proprietorship retail business specializing in premium Apple devices and accessories. The business started as an online paluwagan operation in 2020 and opened its physical showroom on November 8, 2025. The organization was chosen for this study because the owners are willing to participate in the entire project duration, the business has clear problems in inventory and sales tracking that can be addressed by an information system, and the business has an existing customer base and active operations, making it a viable case study.

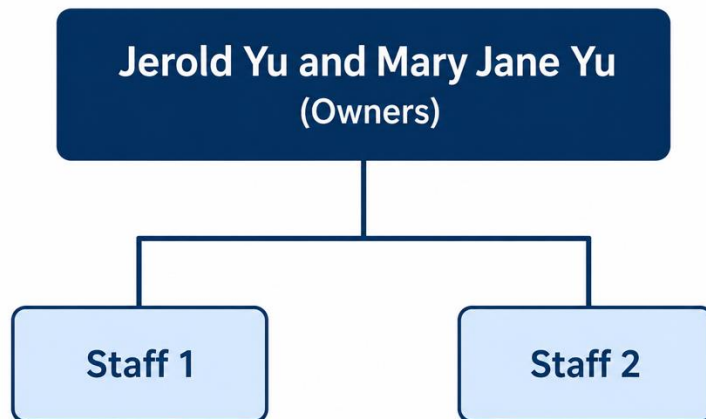


Figure 1.1

Organizational Structure

FLOOR PLAN

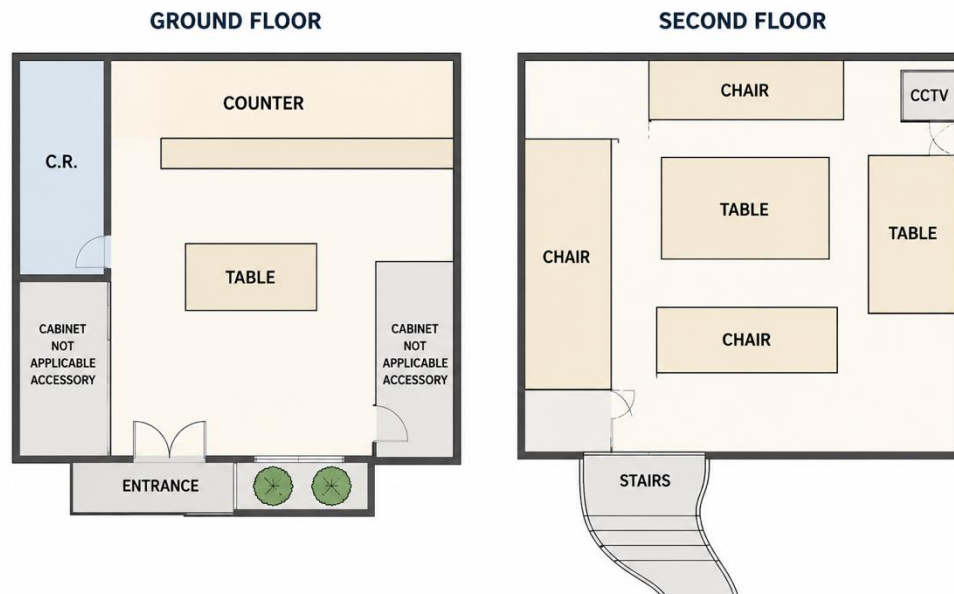


Figure 1.2

Floor Plan

Vision

To become the most trusted and accessible centralized outsourcing platform for local retail shops in Davao City, empowering merchants through technology-driven solutions that provide online visibility, flexible payment tools, and efficient inventory management, while helping customers discover and transact with local businesses easily.

Mission

To provide a centralized digital platform where local retail shops can promote their products and services, manage their inventory and sales through an integrated POS system, and offer flexible payment schemes including paluwagan, layaway, and installments, thereby bridging the gap between offline retail operations and online discoverability while maintaining accurate records and building long-term customer relationships.

Business Environment

Physical Setup. Maharlika Republic operates from a physical showroom located at Door 2, Davao Wisdom Building, F. Torres St., Bajada, Davao City. The showroom is open daily from 10:00

AM to 9:00 PM. Customers can visit to test devices, compare models, and receive diagnostic support. The location is accessible and situated in a commercial area with foot traffic and vehicle access. The business currently has one branch and operates only in Davao City.

ICT Infrastructure. Currently, the organization does not have a dedicated Management Information Systems (MIS) department. They operate with basic ICT resources: desktop computers for Excel-based record keeping, internet connection for social media and customer chat, a credit card terminal for installment payments, a Facebook page and TikTok account (@marexxrepublic) for marketing, and no client-server network or centralized database. The business relies entirely on manual data entry using Microsoft Excel. Multiple staff members edit the same Excel files, leading to version conflicts and data inaccuracies.

Critical Success Factors

For Maharlika Store and its primary partner shop Maharlika Marexx Republic Davao to succeed and grow, the following factors are critical. These attributes have allowed the business to survive competition and will be preserved by the proposed system.

1. **Centralized Platform Visibility** - A user-friendly platform where customers can discover Maharlika Gadgets and other local shops, browse products, and view payment schemes.
2. **Accurate Inventory Management** - Real-time tracking of stock across buy-and-sell, trade-in, and layaway transactions to prevent discrepancies and lost sales.
3. **Unified Sales & Collection Tracking** - A single dashboard that consolidates daily sales, paluwagan draws, layaway payments, GGives, Skyro, and credit card installments.
4. **Customer Communication & Self-Service** - Ability for clients to receive balance and due date updates without contacting shop owners directly.
5. **Efficient Role Delegation** - Clear separation of tasks between owners and staff to reduce bottlenecks during peak hours, enabled by the platform's role-based access.
6. **Standardized Trade-in Policy** - Published guidelines for unit acceptance, condition grading, and valuation, accessible through the platform.
7. **Collection Efficiency** - Automated reminders and a strict collection calendar to minimize overdue accounts.
8. **Scalability for Multiple Shops** - The platform must be designed to onboard additional local retail shops beyond Maharlika Gadgets.

The Current System

Maharlika Marexx Republic Davao currently operates using manual and semi-manual processes with no dedicated software system. The business uses Microsoft Excel for inventory tracking, sales recording, and payment monitoring. No integrated information system (ERP, MRP, TPS, MIS, DSS) is currently in place.

Description of Operations

The following procedures describe the current operations of Maharlika Republic. Operations are handled by four personnel (2 owners + 2 staff) who often perform tasks interchangeably without clear role definitions.

Procedure 1: Customer Inquiry and Product Selection. The customer visits the physical showroom or sends a message via Facebook or TikTok. The staff or owner assists the customer in selecting a device. If the customer is interested in a trade-in, staff evaluates the old device manually and often needs owner approval.

Procedure 2: Sales Transaction. After product selection, staff records the sale in an Excel spreadsheet. Staff manually deducts the sold item from the inventory Excel file. Staff records the payment scheme (cash, layaway, paluwagan, installment, etc.). For

layaway or paluwagan, staff creates a separate Excel sheet or notebook entry for payment tracking.

Procedure 3: Payment Collection. The customer visits the shop to pay. Staff searches through Excel files or notebooks to find the customer's record. Staff manually updates the payment amount and remaining balance. No automated receipt or confirmation is sent to the customer.

Procedure 4: Inventory Restocking. When stock runs low, the owner contacts suppliers. New stock is received and manually entered into the Excel inventory file. No automated low-stock alerts exist.

Procedure 5: Reporting. At the end of the day or week, Jerold Yu and staff manually sum Excel columns. Reports on sales, collections, and best-selling items are manually computed. Reports take hours to prepare and are prone to errors.

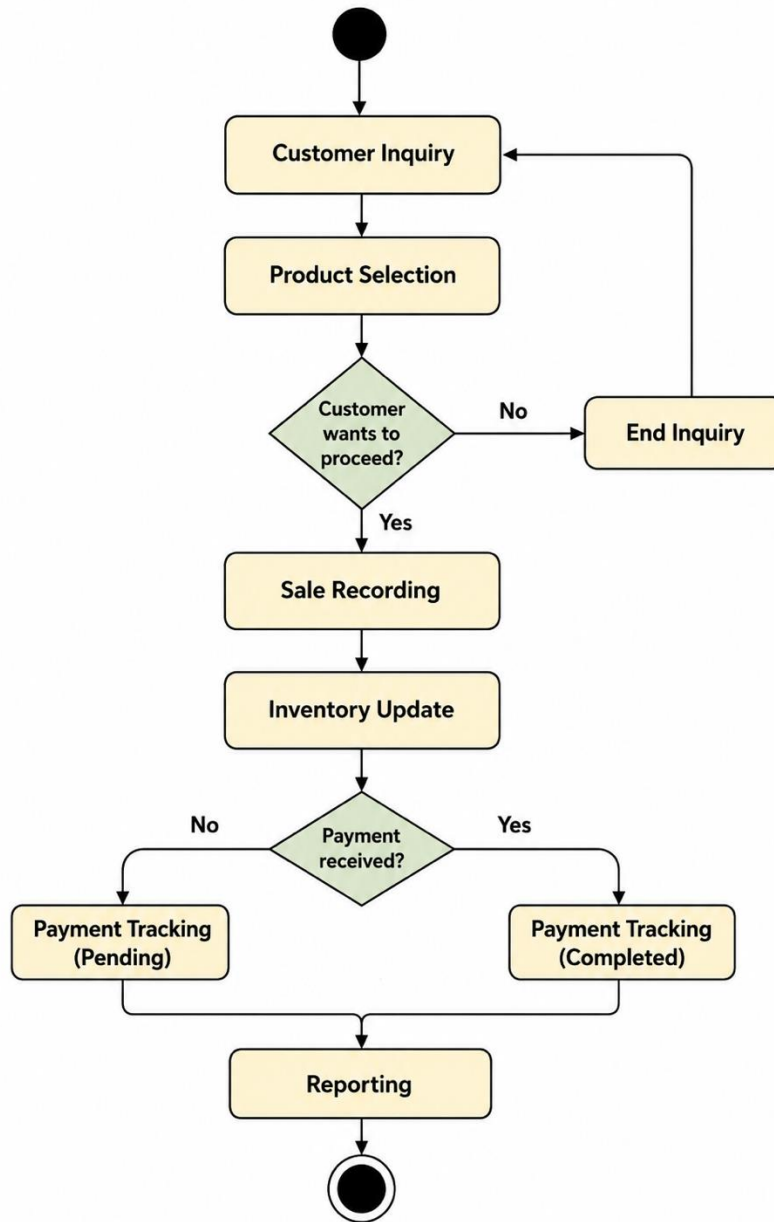


Figure 1.3

Activity of the Current System

Data flows within the current system through manual entry. Customer information, sale details, and payment records are entered into Excel files. Inventory data is updated manually after each sale. Reports are generated by manually summing columns. Data is transformed from raw transaction details into summary information through manual calculation. No automated data transformation exists.

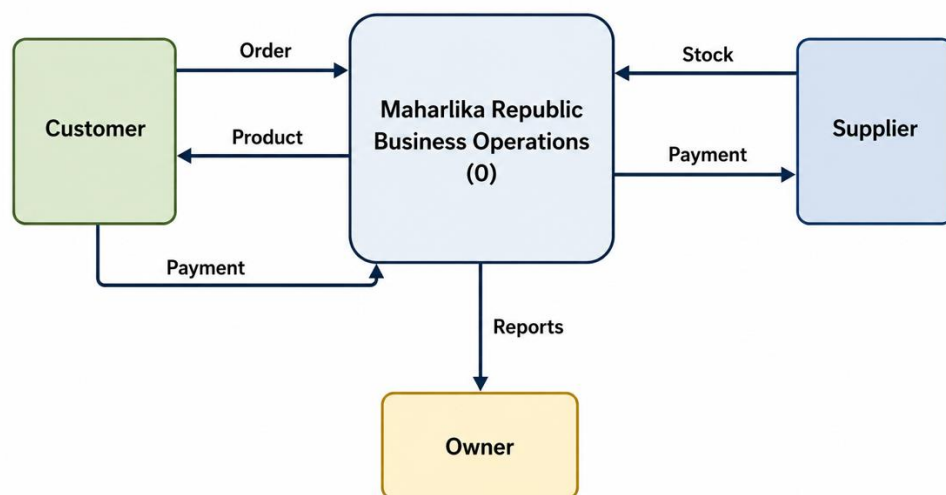


Figure 1.4
Context Level Data Flow Diagram of the Current System

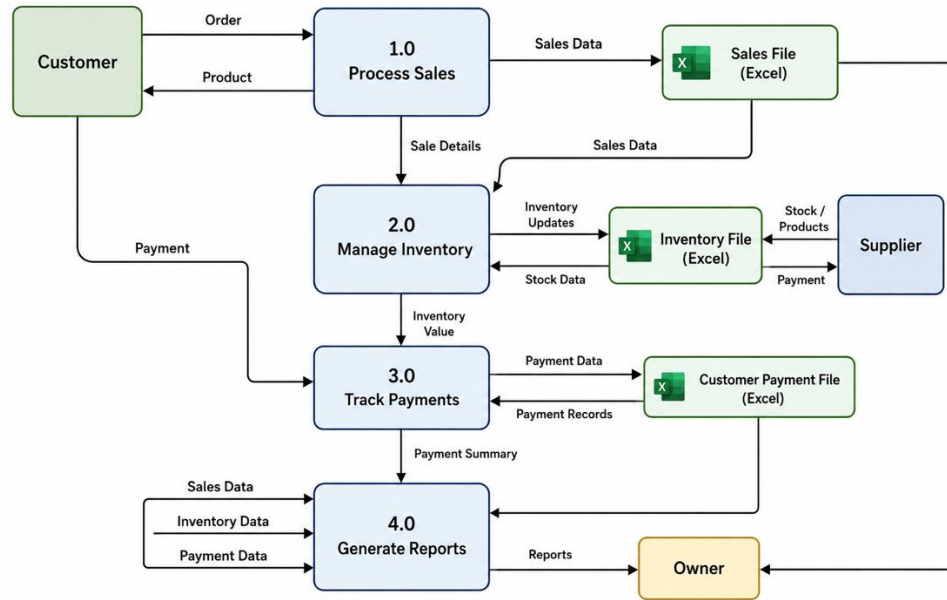


Figure 1.5
Level 0 Data Flow Diagram of the Current System

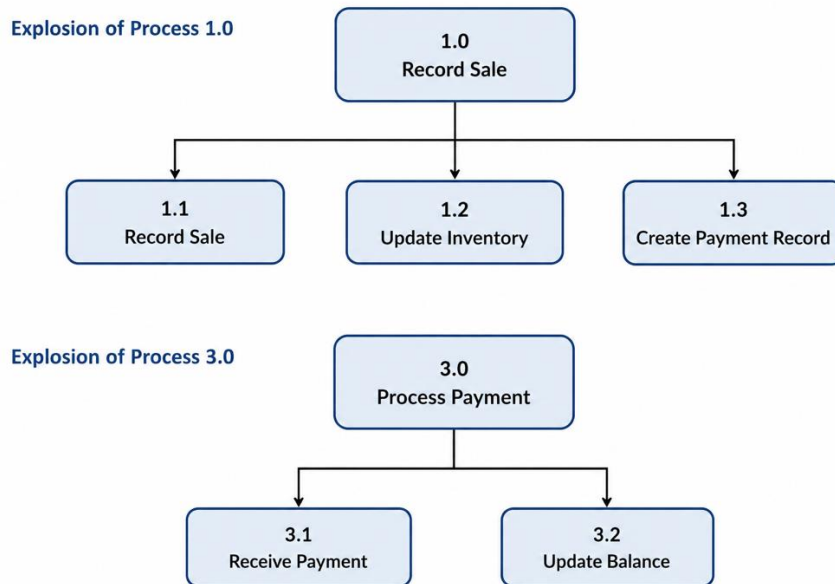


Figure 1.6
DFD Explosion of The Current System

Problem/Opportunity Definition

From the examination of current processes, the following problems and opportunities were identified. The matrix below

enumerates each problem, its corresponding opportunity, cause, and effect.

Table 1.1

Problem - Opportunity Definition Matrix

Key Area	Member 1	Member 2
Performance	Inventory records are difficult to monitor across buy-and-sell, trade-in, and layaway transactions.	Sales performance is difficult to evaluate because records are maintained manually.
Opportunity	Improve inventory visibility and stock monitoring.	Improve sales performance monitoring and reporting accuracy.
Cause	Inventory transactions are recorded in separate files and logs.	Sales records are maintained using manual recording procedures.
Effect	Inventory discrepancies and delayed restocking decisions occur.	Delayed business reporting results in slower decision-making.

Key Area	Member 1	Member 2
Information	Customers have limited access to business information and available services.	Service updates and announcements are difficult to disseminate consistently.
Opportunity	Improve customer access to business information.	Enhance communication with members and clients.
Cause	Business information is distributed through different communication channels.	No centralized communication process exists.
Effect	Reduced visibility and credibility among potential customers.	Missed updates, delayed responses, and lower customer engagement.

Key Area	Member 1	Member 2
----------	----------	----------

Economics	Competitive pricing from online sellers affects profitability.	Delayed collections reduce available operating funds.
Opportunity	Strengthen competitive advantages beyond pricing.	Improve collection monitoring and cash flow management.
Cause	Customers compare prices with large online marketplaces.	Collection tracking relies on manual follow-up procedures.
Effect	Reduced profit margins and customer retention challenges.	Limited funds available for inventory replenishment.

Key Area	Member 1	Member 2
Control	Manual recording increases the risk of data-entry errors.	Customer transactions are recorded separately across services.
Opportunity	Improve record accuracy and consistency.	Strengthen monitoring of customer transaction histories.
Cause	Data is encoded manually without standard validation procedures.	Customer records are maintained in multiple files.
Effect	Financial discrepancies and unreliable records occur.	Duplicate entries and unresolved balance disputes arise.

Key Area	Member 1	Member 2
Efficiency	Staff workload increases during peak transaction periods.	Inventory movement slows during periods of low customer traffic.
Opportunity	Improve task distribution and workflow efficiency.	Increase inventory turnover and sales reach.
Cause	Limited personnel handle multiple services simultaneously.	Sales rely primarily on walk-in customers.
Effect	Longer customer waiting times and operational bottlenecks occur.	Inventory remains unsold for longer periods.

Key Area	Member 1	Member 2
----------	----------	----------

Service	Customers experience delays when requesting balance information.	Customers lack a clear understanding of trade-in requirements.
Opportunity	Improve transparency of customer account information.	Improve customer understanding of trade-in policies.
Cause	Balance inquiries require manual verification by staff.	Trade-in guidelines are not formally communicated.
Effect	Increased inquiry volume and slower response times reduce customer satisfaction.	Customer confusion and valuation disputes may occur.

Performance. The manual inventory management system currently in place at Maharlika Republic has been used since the business started online operations in 2020. Because multiple staff members edit the same Excel files, version conflicts and stock discrepancies occur frequently. If these problems are not solved, the business will continue to experience missed trade-in records, delayed restocking decisions, and reduced sales performance. The opportunity to adopt a cloud-based POS system addresses these issues by automating stock tracking in real time.

Information. The lack of a dedicated website limits the business's online discoverability and professional credibility. Since 2020, the business has relied solely on social media. If this continues, potential customers will not be able to easily find or verify the business, reducing leads and trust. Maharlika Store provides the opportunity to create a centralized promotional platform where shops can be discovered.

Economics. Intense price competition from Lazada and Shopee puts pressure on buy-and-sell margins. However, these online platforms cannot offer the flexible payment schemes that Maharlika Republic provides. If the business does not capitalize on this differentiator, it risks losing price-sensitive customers. The proposed platform highlights these flexible payment options prominently.

Control. Manual data entry without validation leads to human errors that multiply with four people encoding. If unresolved, incorrect inventory counts and financial records will cause customer disputes and unrecorded losses. The opportunity is to create a standardized trade-in pricing guide and form-based data entry within Maharlika Store.

Efficiency. With no formal task division, all four personnel handle all services interchangeably, creating bottlenecks during peak hours. Owners are overburdened with approvals that staff could handle. The proposed platform's role-based access and built-in trade-in guide empower staff to handle standard transactions without owner approval.

Service. Customers cannot check their balances without contacting staff, increasing staff workload and slowing response times. If unresolved, this will reduce client satisfaction and trust. The

opportunity is to send automated SMS or chat balance updates through Maharlika Store.

Functional Requirements

Based on the identified problems and opportunities, Maharlika Republic needs a combination of a **Transaction Processing System (TPS)** for sales and payments, and a **Management Information System (MIS)** for reporting. The Maharlika Store platform will provide the following functional requirements. All items written here are based on interviews conducted with the system owners (Jerold and Mary Jane Yu) and system users (Staff 1 and Staff 2).

A. Customer-Facing Storefront

1. **Browse Shops** - The system shall allow customers to view shop profiles, products, and payment schemes.
 - *Features:* Homepage with featured shops, search bar, category filters.
 - *Data needed:* Shop name, address, contact number, product images, prices.
2. **View Product Catalog** - The system shall display products with images, prices, and availability status.
 - *Features:* Product grid, quick view, stock status indicator.

- *Data needed:* Product name, description, price, stock quantity, image URL.
3. **Contact Shop** - The system shall provide contact information and directions to the physical showroom.
- *Features:* Clickable phone number, map embed, contact form.
 - *Data needed:* Phone number, email, address, Google Maps link.

B. Inventory Management

1. **Real-Time Stock Tracking** - The system shall automatically deduct inventory when a sale is recorded.
- *Features:* Auto-update on sale, manual adjustment option.
 - *Data needed:* Product ID, quantity on hand, reorder point.
2. **Low-Stock Alerts** - The system shall notify owners when stock falls below a defined threshold.
- *Features:* Dashboard notification, email alert (optional).
 - *Data needed:* Product ID, current stock, threshold value.
3. **Add/Edit Products** - The system shall allow owners to add new products, update prices, and remove discontinued items.
- *Features:* Product form with validation, image upload.

- *Data needed:* Product name, category, price, stock, image.

4. **Trade-In Guide** - The system shall provide a standardized valuation guide for trade-in transactions.

- *Features:* Searchable guide, auto-calculation of trade-in value.
- *Data needed:* Device model, condition grade, base value.

C. Sales and POS

1. **Record Sale** - The system shall record sales with customer information, items sold, total amount, and payment scheme.

- *Features:* Customer search, item selection, auto-total calculation.
- *Data needed:* Customer ID, product IDs, quantities, payment scheme.

2. **Multi-Payment Support** - The system shall support cash, layaway, paluwagan, credit card, Skyro, Salmon, GGives, Maya, and COD.

- *Features:* Payment scheme selection, terms configuration.
- *Data needed:* Payment scheme type, down payment (if any), number of installments.

3. Trade-In Processing - The system shall allow staff to process standard trade-ins without owner approval; non-standard trades require owner approval.

- *Features:* Trade-in form, value calculation, approval workflow.
- *Data needed:* Trade-in device details, condition, calculated value.

D. Customer and Payment Management

1. Customer Profile - The system shall maintain customer profiles with contact information and transaction history.

- *Features:* Add/edit customer, view transaction history.
- *Data needed:* Name, contact number, email, join date.

2. Consolidated Balance - The system shall show all active balances (layaway, paluwagan, installment) for a single customer.

- *Features:* Balance summary dashboard by customer.
- *Data needed:* Customer ID, balance by scheme, total balance.

3. Record Payment - The system shall record payments and automatically update remaining balances.

- *Features:* Payment entry form, auto-balance update.

- *Data needed:* Customer ID, amount paid, payment date, reference transaction.
4. **Payment Reminders** - The system shall allow staff to send balance updates via SMS or chat.
- *Features:* "Send Reminder" button, pre-formatted message.
 - *Data needed:* Customer contact number, current balance, due date.

E. Reporting

1. **Daily Sales Report** - The system shall generate a report of all sales for a selected date.
- *Features:* Date picker, export to Excel/PDF.
 - *Data needed:* Sale date, total amount, payment schemes, staff member.
2. **Weekly Sales Report** - The system shall generate a summary of sales by day, service type, and staff member.
- *Features:* Weekly summary view, performance metrics.
 - *Data needed:* Sale dates, daily totals, staff IDs.
3. **Collection Report** - The system shall list overdue accounts and pending collections.
- *Features:* Overdue list, task assignment.

- *Data needed:* Customer name, balance, due date, assigned staff.
4. **Inventory Report** - The system shall show current stock levels and low-stock items.
- *Features:* Current stock view, low-stock filter.
 - *Data needed:* Product name, stock quantity, threshold.

F. User Management

1. **Role-Based Access** - The system shall support two roles: Owner (full access) and Staff (limited access).
- *Features:* Role assignment, permission restrictions.
 - *Data needed:* User ID, role type.
2. **Audit Log** - The system shall record which user performed each transaction.
- *Features:* Transaction log with user ID and timestamp.
 - *Data needed:* User ID, action, timestamp, record affected.

Use Cases

Table 1.2

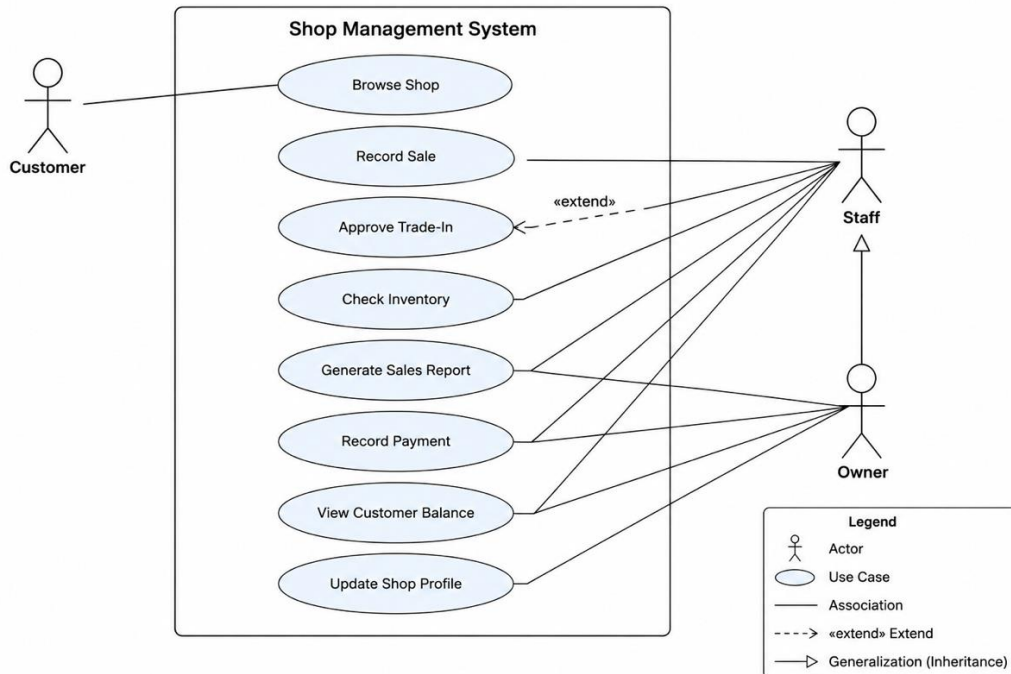
Use Case Glossary

Use-Case Name	Use-Case Description	Participating Actors and Roles
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Browse Shop	Customer views shop profile and products on Maharlika Store	Customer
Record Sale	Staff records a sale transaction	Staff, Owner
Approve Trade-In	Owner approves a non-standard trade-in request	Owner
Check Inventory	Staff views current stock levels	Staff, Owner
Generate Sales Report	Owner generates daily/weekly sales report	Owner
Record Payment	Staff records customer payment	Staff, Owner
View Customer Balance	Staff views customer's outstanding balances	Staff, Owner
Update Shop Profile	Owner updates shop information on the storefront	Owner

Figure 1.7

Use Case Diagram



Feasible Alternatives

The following alternative solutions were considered to address the identified problems. Each alternative is described below.

Alternative 1: Continue with Excel (Do Nothing). Maintain current manual processes using Excel spreadsheets and notebooks. No changes to operations. This alternative costs nothing but does not solve any problems.

Alternative 2: Off-the-Shelf POS System + Separate Website. Purchase a commercial POS system (e.g., Loyverse, Toast) and build a separate promotional website. This provides proven POS

functionality but may not support paluwagan and layaway schemes. It also requires integration between two separate systems.

Alternative 3: Maharlika Store Platform (Proposed). Build a custom web-based platform with customer-facing storefront and integrated POS/inventory dashboard for shop owners. This alternative is fully customizable, supports all payment schemes, and scales for multiple shops.

Alternative 4: Hybrid (Google Forms + Sheets + Social Media). Use Google Forms for data entry, Google Sheets as database, and Facebook for promotion. This is low cost but lacks real-time inventory and has version control issues.

Table 1.3

Detailed Budget Computations for Candidate 2 (Attachment B)

Item	Cost per Month	Annual Cost
POS Subscription (4 users × ₱500)	₱2,000	₱24,000
Website Hosting	₱500	₱6,000
Domain Name	₱0 (included)	₱0
Total	₱2,500	₱30,000

Table 1.4

Detailed Budget Computations for Candidate 3 (Attachment B)

Item	One-time Cost	Monthly Cost

Development (Capstone Project)	₱0 (academic requirement)	₱0
Website Hosting (Vercel - Free Tier)	₱0	₱0
Database Hosting (Supabase - Free Tier)	₱0	₱0
Domain Name (if needed)	₱0 (using vercel.app subdomain)	₱0
Total	₱0	₱0

Feasibility Analysis

Operational Feasibility

Alternative 1 is currently in use but has very low operational effectiveness due to multi-user chaos. Alternative 2 requires staff to learn two separate systems. Alternative 3 offers high operational feasibility because the system is tailored to the exact workflows of Maharlika Republic. Alternative 4 requires staff to manage multiple Google tools manually.

Technological Feasibility

Alternatives 1 and 4 use existing or free technologies. Alternative 2 requires POS hardware and vendor support. Alternative 3 requires web development skills, which the capstone team possesses, and can be hosted on affordable cloud platforms.

Economic Feasibility

Alternative 1 has no direct cost but incurs hidden costs from errors and lost sales. Alternative 2 requires monthly subscriptions (approximately ₱1,500-3,000 per month for 4 users) plus website hosting. Alternative 3 requires one-time development (covered by the capstone project) and hosting (approximately ₱500-1,000 per month). Alternative 4 is free for basic use.

Schedule Feasibility

Alternative 1 is immediate. Alternative 2 can be set up in 1-2 weeks. Alternative 3 requires 8-12 weeks for a prototype, which fits within the capstone timeline. Alternative 4 can be set up in 1-2 weeks.

Table 1.5

Feasibility Analysis Matrix

Feasibility Dimension	A1: Continue with Excel (Do Nothing)	A2: Off-the-Shelf POS + Separate Website	A3: Maharlika Store Platform (Proposed)	A4: Hybrid (Forms + Sheets + Social Media)
Operational Feasibility	Very Low - Multi-user chaos, version	Medium - Staff must learn two separate	High - System tailored to exact	Low - Staff must manage multiple

	conflicts, no clear roles	systems; workflows may not match	workflows; role-based access designed for 4 personnel	Google tools manually; no unified system
Technological Feasibility	High - Uses existing Excel; no new technology needed	High - Vendor provides POS hardware and software support	Medium - Requires web development skills (capstone team has this); cloud hosting available	High - Uses free Google tools; no special hardware required
Economic Feasibility	High - No direct cost, but hidden costs from errors and lost sales	Low - Monthly subscription (₱1,500- 3,000 × 4 users) + website hosting (₱500- 1,000/month)	Medium - One-time development (capstone project); hosting only (₱500- 1,000/month)	High - Free for basic use; minimal to no cost
Schedule Feasibility	Immediate - No development needed	Short to Medium - 1- 2 weeks for setup and training	Medium - 8- 12 weeks for prototype; fits within	Short - 1- 2 weeks for setup

			capstone timeline	
Risk Level	High - Errors, conflicts, no online visibility	Medium - Integration issues, feature mismatch, ongoing subscription costs	Low - Full control over features; tailored exactly to needs	Medium - Scalability limits; version conflicts still possible

Recommendation

Alternative 3 (Maharlika Store Platform) is recommended because it fully supports all unique payment schemes, provides both customer-facing promotion and internal management in one unified platform, can be built within the capstone project timeline, and allows future scalability to onboard other local retail shops.

Scope and Limitation

The Maharlika Store platform is designed to support the daily operations of the business through a web-based management system. Its scope includes a customer-facing website where users can browse shop information, products, payment schemes, and contact details without performing online purchases. The system provides inventory management with real-time stock monitoring, low-stock alerts, and trade-in support, as well as sales and point-of-sale (POS)

recording for various payment methods such as cash, layaway, paluwagan, GGives, Skyro, and credit cards. Additional features include customer balance management, payment tracking with automatic balance updates, sales reporting, role-based access for owners and staff, a trade-in pricing guide, and audit logs for transaction monitoring.

However, several functionalities are outside the scope of the current capstone project. These include full e-commerce checkout, online payment processing, direct integration with third-party financing services such as GGives, Skyro, and Salmon, customer mobile applications, automated SMS gateway subscriptions, BIR or tax reporting integration, multi-branch management, full accounting and general ledger features, advanced security mechanisms such as biometrics, and the onboarding of additional shops. While the system database is designed to support future expansion, the current implementation is focused solely on the Bajada showroom and its operational requirements.

The Proposed System

General Objective

To design and develop Maharlika Store, a centralized outsourcing and promotional platform for local retail shops, with an integrated cloud-based Point-of-Sale and inventory management

system for shop owners. The platform will serve two primary audiences: (1) customers who can browse shops, view products and payment schemes, and discover local retail shops like Maharlika Gadgets; and (2) shop owners and staff who can manage inventory, record sales, track collections across all payment schemes (buy-and-sell, trade, paluwagan, layaway, GGives, Skyro, credit card installment), generate reports, and communicate with customers. The platform specifically addresses the needs of Maharlika Marexx Republic Davao (Maharlika Gadgets) as the primary partner shop, with the capability to onboard additional local retail shops in the future.

Specific Objectives

Based on the problem/opportunity definition and the general objective, the following specific outcomes need to be accomplished:

1. Provide a customer-facing storefront where Maharlika Gadgets and other local shops can be discovered, viewed, and contacted by potential customers.
2. Reduce stock discrepancies by at least 90% through real-time multi-user inventory updates within the shop owner dashboard.
3. Generate daily and weekly sales reports in under 1 minute, including staff performance metrics.

4. Provide a single customer view showing all active balances across layaway, installment, and paluwagan.
5. Enable staff to record standard transactions (non-trade-in) in under 30 seconds.
6. Reduce owner workload by 50% by allowing staff to handle standard trade-ins using the built-in pricing guide.
7. Implement role-based access so owners can delegate tasks without losing control over approvals.
8. Create a scalable platform architecture that allows onboarding of additional local retail shops in the future.

Scope of Automation

The following manual processes will be automated by the Maharlika Store platform. The limitations are clearly defined by constructing the DFD and activity diagram of the new system

Table 1.6

Scope of Automation

Manual Process	Automation Feature
No online presence for the shop	Customer-facing storefront on Maharlika Store
Multiple staff editing the same Excel file	Cloud database with real-time updates and conflict prevention
Recording stock in Excel after each sale	Real-time inventory deduction upon sale
Manually summing Excel columns for daily sales	Dashboard with auto-generated daily/weekly sales totals

Tracking payments in separate files	Unified customer record with all balances and history
Manually checking low-stock items	Low-stock alert dashboard
Encoding data without validation	Form-based input with dropdowns and validation
Staff calling owner for every trade-in quote	Built-in trade-in pricing guide; staff can quote standard items
Sending individual balance inquiries via chat	"Send Reminder" button that copies balance to chat/SMS
Owners manually assigning collection follow-ups	System-generated collection calendar with task assignment
No record of who did what transaction	Audit log showing which staff member recorded each transaction
Manual updating of shop promotions and products	Shop owner dashboard to update shop profile instantly

Processes that remain manual (but supported by the platform) :

- Physical inventory counting for audit (system provides expected counts)
- Customer negotiation beyond standard trade-in guide (system records final value after owner approval)
- Handling cash and payment collection (system records after collection)
- Sending actual SMS/chat messages (system prepares the message content)
- Final customer purchase transaction (customers contact shop directly; no online checkout)

Figure 1.8

Activity Diagram of The Proposed System

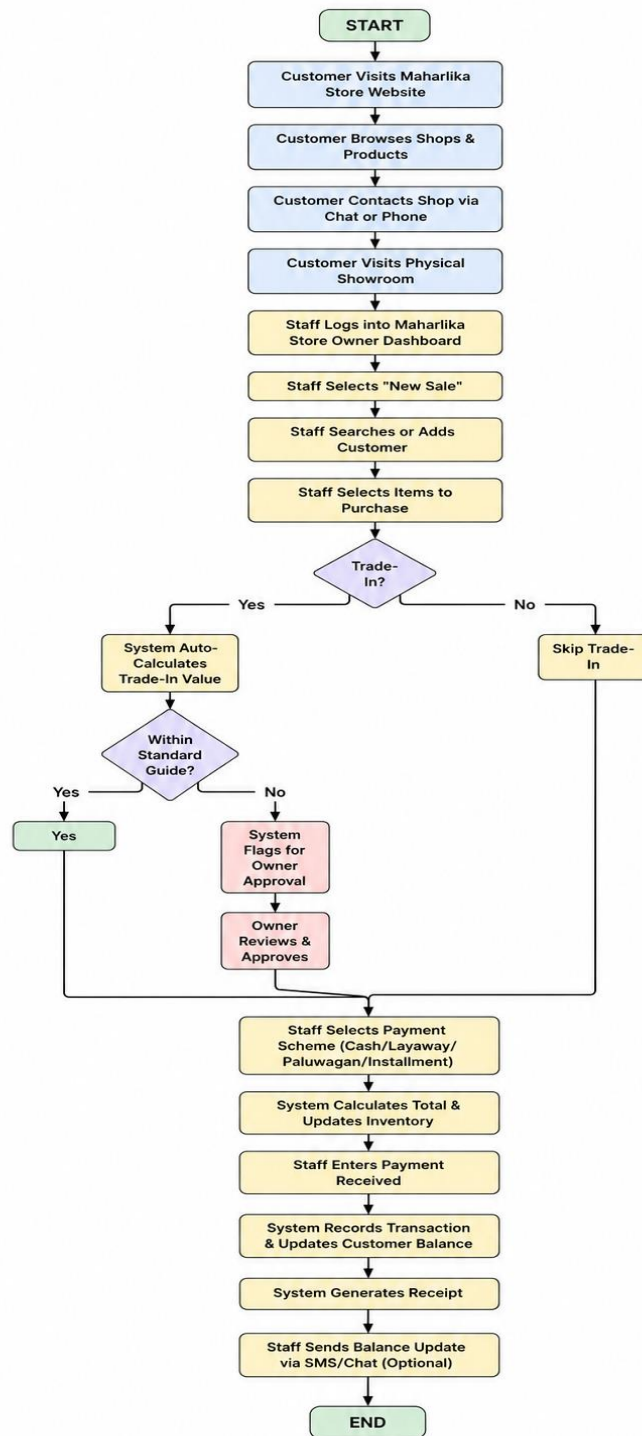


Figure 1.9

Context Level Data Flow of The Proposed System

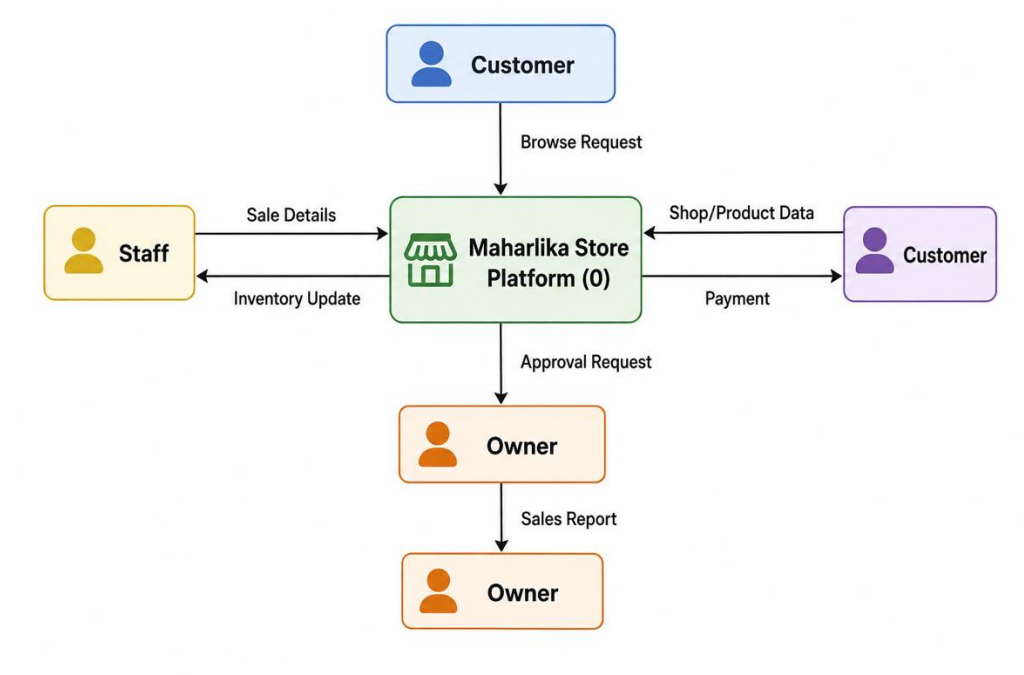


Figure 1.10

Level 0 Data Flow Diagram of the Proposed System

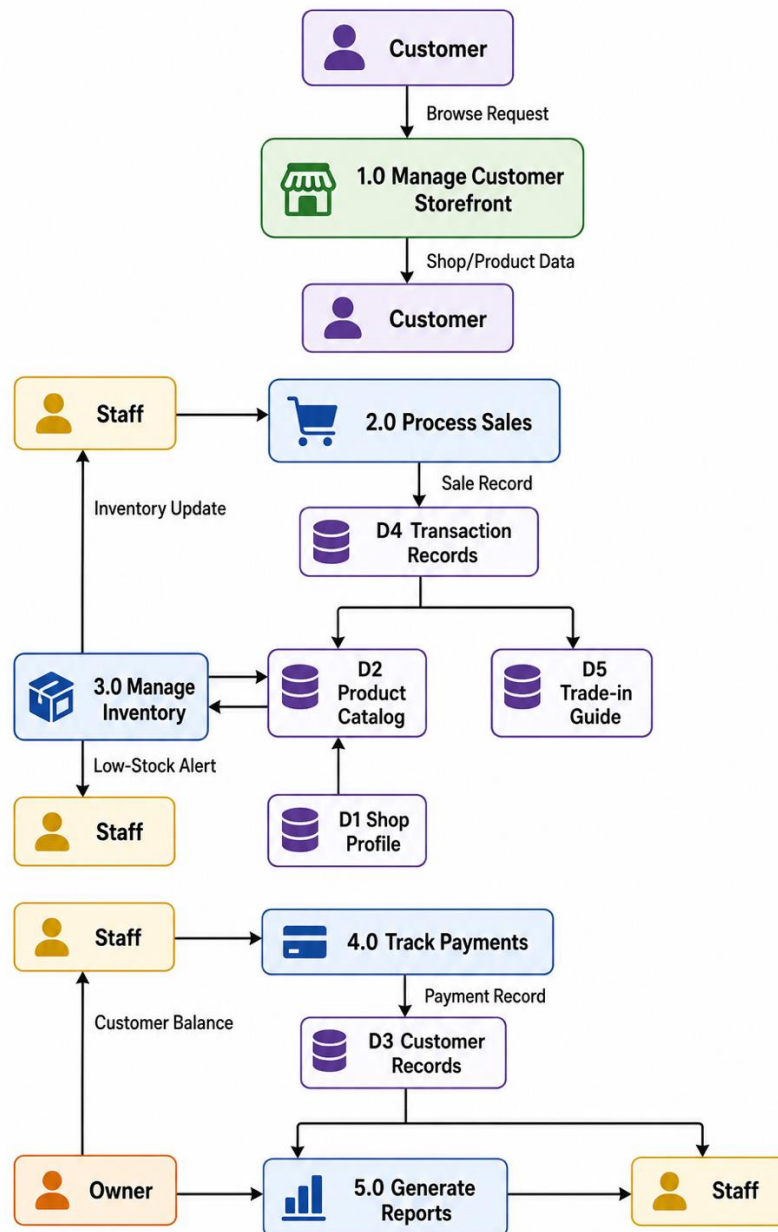


Figure 1.11

DFD Explosion of the Proposed System

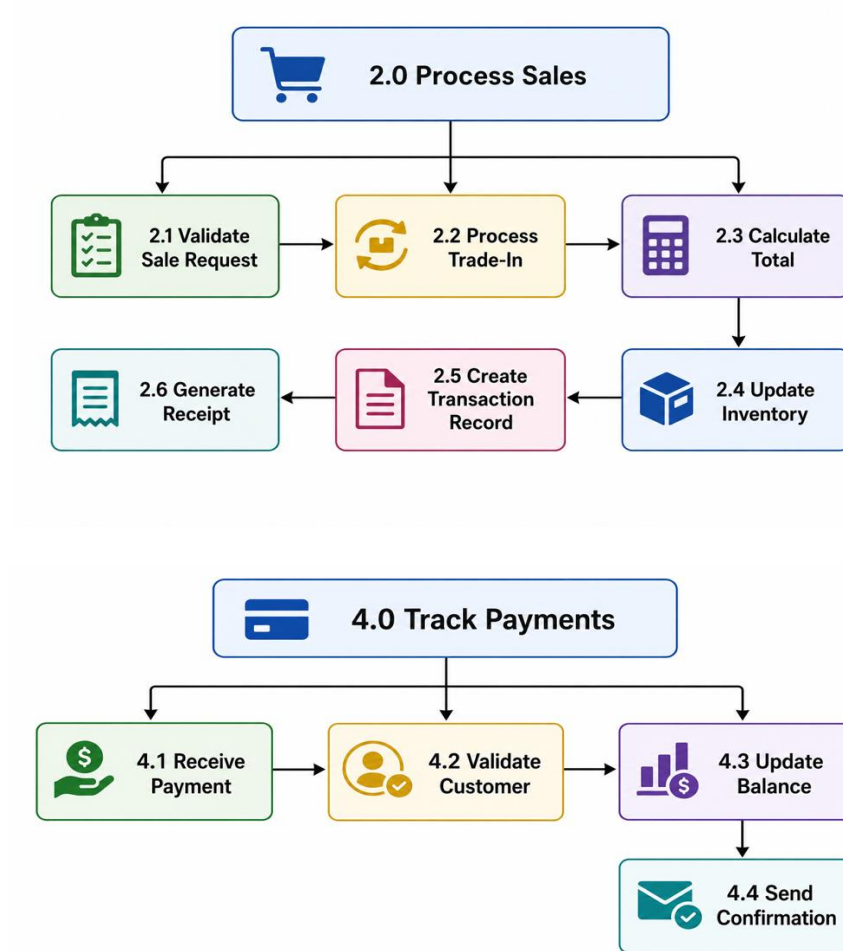


Figure 1.12

Entity Relationship Diagram of the Proposed System

Figure 1.12: Entity Relationship Diagram (Conceptual ERD)

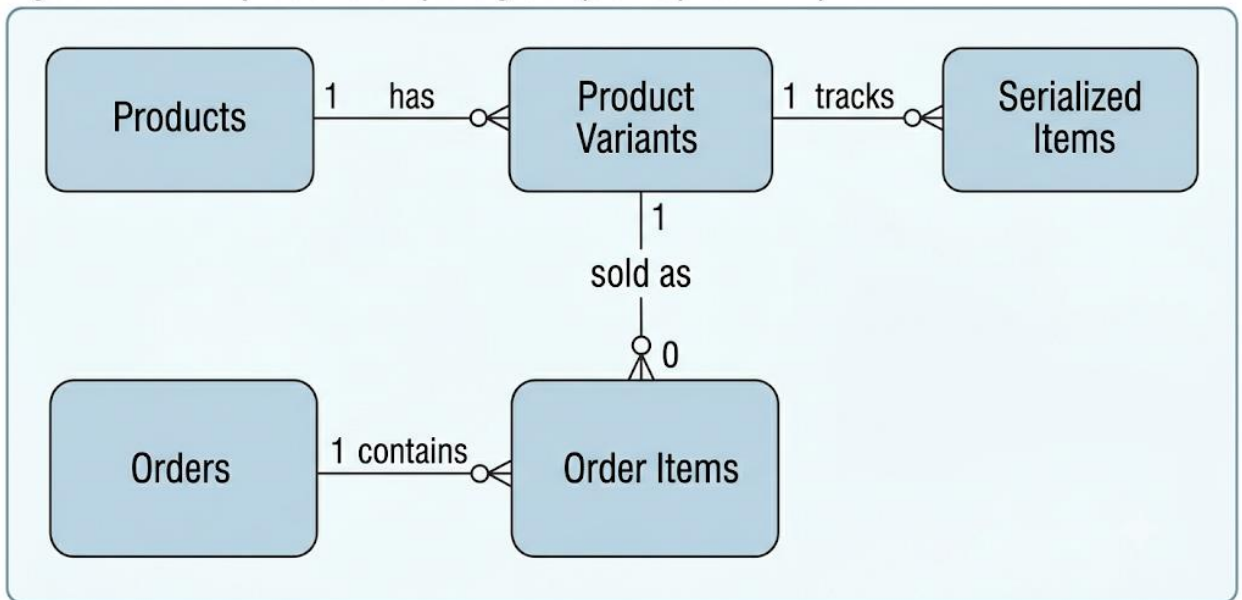


Figure 1.13

Entity Relationship Diagram of the Proposed System (Key-Based ERD)

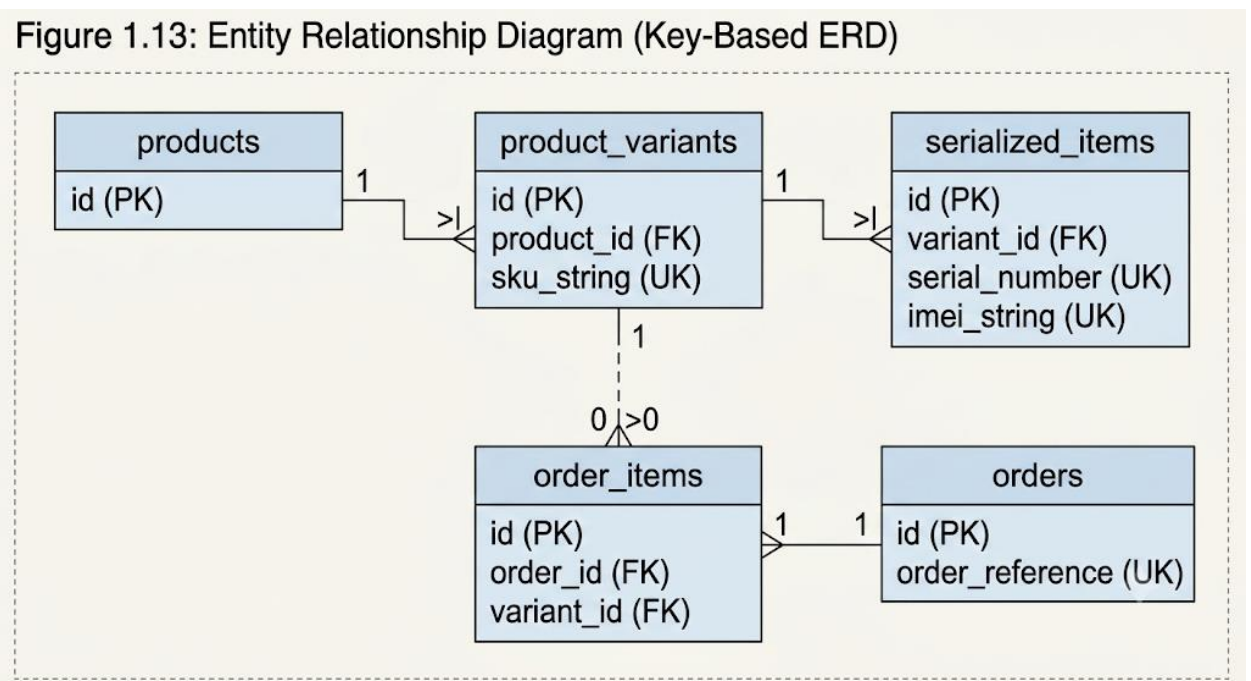
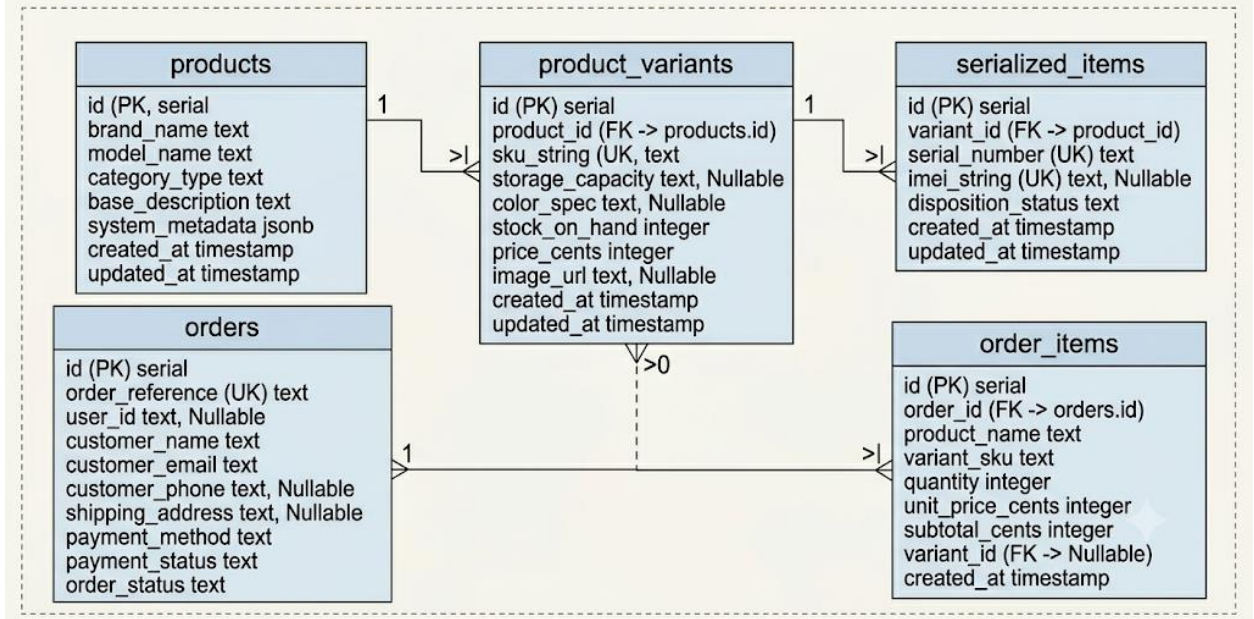


Figure 1.14

Entity Relationship Diagram of the Proposed System (Fully Attributed ERD)

Figure 1.14: Entity Relationship Diagram (Fully Attributed ERD)



Data Dictionary

I. Data Dictionary for the Data Flow Diagram

Data Flow Name	Description	Composition
Browse Request	Customer request to view shops/products	Customer_ID + Request_Type + Filters
Shop/Product Data	Shop and product information displayed to customer	Shop_Name + Address + Contact + Product_Name + Price + Stock_Status
Sale Details	Information about a sales transaction	Customer_ID + Product_ID + Quantity + Payment_Scheme + TradeIn_Details
Inventory Update	Stock adjustment after sale	Product_ID + Quantity_Deducted

Transaction Record	Record of completed sale	Sale_ID + Customer_ID + Total_Amount + Payment_Scheme + Date
Payment	Customer payment information	Customer_ID + Amount + Payment_Method + Reference_Transaction
Customer Balance	Current outstanding balance for a customer	Customer_ID + Customer_Name + Layaway_Balance + Paluwagan_Balance + Installment_Balance
Approval Request	Trade-in approval request sent to owner	TradeIn_ID + Customer_ID + Product_Traded + Proposed_Value
Sales Report	Summary of sales for a period	Date_Range + Total_Sales + Sales_By_Staff + Sales_By_Service
Low-Stock Alert	Notification for products below threshold	Product_ID + Product_Name + Current_Stock + Threshold

II. Data Dictionary for the Entity Relationship

Diagram

5. Entity	Business Definition	Attributes
Shop	The retail business (Maharlika Republic) that sells products and offers payment schemes	shop_id (PK), business_name, address, contact_number, email, logo_url, description, operating_hours
User	A person who operates the	user_id (PK), shop_id (FK), full_name, role (Owner/Staff),

	system (Owner or Staff)	username, password_hash, created_at
Product	A device or accessory offered for sale	product_id (PK), shop_id (FK), name, category, price, stock_quantity, low_stock_threshold, image_url, is_active, trade_in_value_guide
Customer	An individual who purchases items or avails payment schemes	customer_id (PK), first_name, last_name, contact_number, email, join_date
Sale	A transaction where a customer purchases one or more products	sale_id (PK), customer_id (FK), user_id (FK), sale_date, total_amount, payment_scheme, status
Sale_Item	Line item within a sale, linking a product to a sale	sale_item_id (PK), sale_id (FK), product_id (FK), quantity, unit_price, subtotal
Payment	A monetary transaction where a customer pays toward a balance	payment_id (PK), customer_id (FK), sale_id (FK), layaway_id (FK), paluwagan_id (FK), amount, payment_date, payment_method, remaining_balance
Layaway	A payment plan where a customer reserves a product and pays over time	layaway_id (PK), customer_id (FK), product_id (FK), total_price, amount_paid, balance, start_date, due_date, status
Paluwagan	A community savings plan where members contribute and draw a prize	paluwagan_id (PK), customer_id (FK), contribution_amount, total_contributions, draw_date, status

TradeIn	A transaction where a customer exchanges an old device for credit	tradein_id (PK), customer_id (FK), product_id (FK), trade_in_value, condition_grade, approved_by (FK), approval_status
Audit_Log	A record of user actions for security and accountability	log_id (PK), user_id (FK), action, record_type, record_id, timestamp

Part II

Project Description

Project Title

Maharlika Store: A Centralized Outsourcing and Promotional Platform for Local Retail Shops

The title "Maharlika Store" was chosen to reflect the brand identity of the primary partner shop, Maharlika Marexx Republic Davao (operating as Maharlika Gadgets). The word "Maharlika" is a Filipino term that means "noble" or "elite," which aligns with the business's focus on premium Apple devices. The word "Store" indicates a centralized marketplace where multiple shops can be listed, discovered, and promoted to customers, similar to the iTunes Store or App Store model.

The title was evaluated and approved by the instructor based on the following criteria:

1. It reflects the type of information system being proposed (centralized outsourcing and promotional platform).
2. It includes the name of the primary partner company (Maharlika).
3. It clearly describes the purpose of the system.

Project Logo

Figure 2.1

Project Logo



The logo consists of a stylized shop icon combined with the text "Maharlika Store." The color scheme uses gold to represent trust, quality, and Filipino heritage. The tagline "Discover Local Shops, One Platform" communicates the platform's value proposition to customers.

Project Organization

The project team is composed of two members, each with specific roles and responsibilities based on their strength and weaknesses. The team follows a hierarchal structure similar to industry settings.

Figure 2.2

Project Organizational Structure



This Structure follows the industry setting where each member has a primary role while still collaborating across tasks. The instructor assigned the grouping based on the strengths and weaknesses of each member, as required by the SAD Guidelines.

Project Methodology

The project will follow the **Agile Scrum methodology**. This process model was chosen because it allows for iterative development, frequent client feedback, and flexibility in adapting to changing requirements. The capstone project timeline of 8-12 weeks for development fits well with Scrum's sprint structure.

Figure 2.3

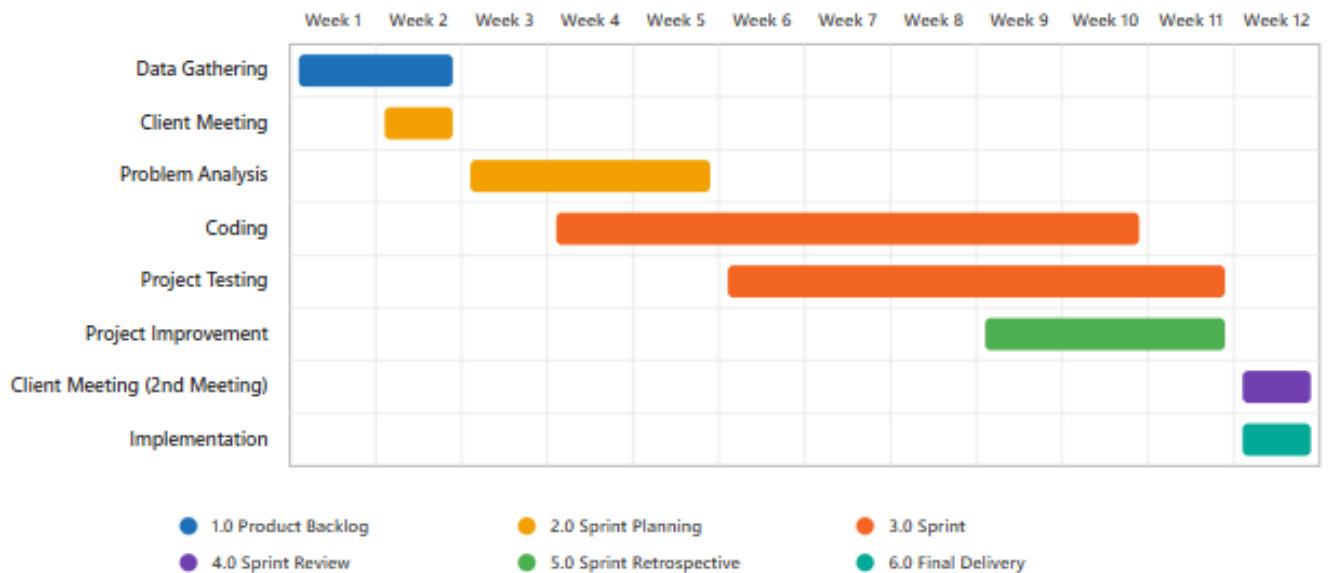
Project Methodology (Agile Scrum Process)



Project Schedule

The project schedule is divided into two parts: (1) a Gantt chart showing the overall timeline, and (2) a detailed timeline indicating the person assigned for each activity.

Figure 2.4 Gantt Chart



Technology

The following technologies will be used to develop the Maharlika Store platform. These technologies were chosen based on their suitability for the project requirements, the team's skill set, and cost-effectiveness.

Table 2.1

Technology Stack

Category	Technology	Purpose
Frontend Framework	React.js or HTML/CSS/JS	Build responsive customer-facing storefront and admin dashboard
Backend Framework	Node.js with Express or PHP Laravel	Handle server-side logic, API endpoints, and business rules
Database	PostgreSQL or MySQL	Store shop profiles, products, customers, sales, and payment records
Hosting	Vercel (frontend) + Supabase (backend/database)	Deploy and host the application on the cloud
Authentication	JWT (JSON Web Tokens)	Implement role-based access control for owners and staff
Version Control	Git + GitHub	Track code changes and enable team collaboration
Project Management	Trello or Jira	Manage sprints, tasks, and progress tracking
Documentation	Google Docs / Microsoft Word	Create and maintain project documentation

Diagramming	Miro	Create DFDs, ERDs, use case diagrams, and activity diagrams
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Part III

USER ACCEPTANCE AND TRAINING

User Acceptance Testing

To verify that the proposal has been implemented correctly, testing and training were administered to the users in the organization. The testing was conducted in two types:

Preliminary Testing and Parallel Testing.

Preliminary Testing

was done internally by the team members only. The team tested the system for bugs, errors, and functionality before presenting it to the client.

Parallel Testing

was done with the users. This testing simulated the actual operations within the organization while using the new system. The users affixed their signature to indicate that they were satisfied with the system. The parallel testing was conducted after training had been successfully administered.

For each of the items specified in the functional requirements, test cases were written and the results were documented. This shows whether the team complied with the functional requirements and that they are acceptable to the client.

Table 3.1

User Acceptance Testing (UAT) Results

Test Case #	Tested By	Tested Date	Expected Outcome	Actual Outcome	Remarks
UAT-01	Jerold Yu (Owner)	June 6, 2026	Be able to login as Owner and access all features	Successfully logged in; all features accessible	No problems found
UAT-02	Mary Jane Yu (Owner)	June 6, 2026	Be able to view dashboard and reports	Dashboard displayed correctly; reports generated	No problems found
UAT-03	Staff 1	June 6, 2026	Be able to login as Staff and record a sale	Successfully recorded sale; inventory updated	No problems found
UAT-04	Staff 2	June 6, 2026	Be able to record	Payment recorded;	No problems found

			customer payment	balance updated correctly	
UAT-05	Staff 1	June 6, 2026	Be able to process standard trade-in	Trade-in value auto-calculated; no owner approval needed	No problems found
UAT-06	Jerold Yu (Owner)	June 6, 2026	Be able to approve non-standard trade-in	Approval request received; approved successfully	No problems found
UAT-07	Mary Jane Yu (Owner)	June 6, 2026	Be able to generate daily sales report	Report generated in under 1 minute; data accurate	No problems found
UAT-08	Staff 2	June 6, 2026	Be able to check customer balance	Customer balance displayed with all active schemes	No problems found
UAT-09	Staff 1	June 6, 2026	Be able to send payment reminder	Reminder message prepared; copied to chat	No problems found
UAT-10	Jerold Yu (Owner)	June 6, 2026	Be able to update shop profile		

Parallel Testing Result

Parallel testing was conducted after training was successfully administered. The system ran simultaneously with the existing manual processes (Excel) for 3 days to verify accuracy.

Table 3.2

User Acceptance Testing Result

Test Case #	Tested By	Tested Date	Expected Outcome	Actual Outcome	Remarks
UAT-01	Jerold Yu (Owner)	June 6, 2026	Be able to login as Owner and access all features	Successfully logged in; all features accessible	No problems found
UAT-02	Mary Jane Yu (Owner)	June 6, 2026	Be able to view dashboard and reports	Dashboard displayed correctly; reports generated	No problems found
UAT-03	Staff 1	June 6, 2026	Be able to login as Staff and record a sale	Successfully recorded sale; inventory updated	No problems found
UAT-04	Staff 2	June 6, 2026	Be able to record customer payment	Payment recorded; balance updated correctly	No problems found
UAT-05	Staff 1	June 6, 2026	Be able to process standard trade-in	Trade-in value auto-calculated; no owner approval needed	No problems found
UAT-06	Jerold Yu (Owner)	June 6, 2026	Be able to approve non-standard trade-in	Approval request received; approved successfully	No problems found

UAT-07	Mary Jane Yu (Owner)	June 6, 2026	Be able to generate daily sales report	Report generated in under 1 minute; data accurate	No problems found
UAT-08	Staff 2	June 6, 2026	Be able to check customer balance	Customer balance displayed with all active schemes	No problems found
UAT-09	Staff 1	June 6, 2026	Be able to send payment reminder	Reminder message prepared; copied to chat	No problems found
UAT-10	Jerold Yu (Owner)	June 6, 2026	Be able to update shop profile	Shop profile updated; changes reflected on storefront	No problems found

All 10 test cases passed. The users expressed satisfaction with the system. No major issues or bugs were reported. Minor suggestions for future enhancements were noted (see Recommendations in Part IV).

Parallel Testing Result

Parallel testing was conducted after training was successfully administered. The system ran simultaneously with the existing manual processes (Excel) for 3 days to verify accuracy.

Table 3.3

Parallel Testing Result

Test Date	Transaction Type	Manual Result (Excel)	System Result (Maharlika Store)	Match?
June 6, 2026	Sale (Cash)	₱46,990	₱46,990	Yes
June 6, 2026	Layaway Payment	₱5,000	₱5,000	Yes
June 6, 2026	Trade-In (Standard)	₱8,500	₱8,500	Yes
June 6, 2026	Paluwagan Contribution	₱2,000	₱2,000	Yes
June 6, 2026	Installment Payment	₱3,500	₱3,500	Yes
June 6, 2026	Inventory Update	-5 units	-5 units	Yes

All transactions matched between the manual Excel records and the Maharlika Store system. No discrepancies were found. The system proved to be as accurate as the manual process, with the added benefit of real-time updates and automated reporting.

UAT Sign-off

The following users have affixed their signature to confirm that they are satisfied with the system and that the system meets the functional requirements.

Table 3.4

UAT Sign-off

Name	Role	Signature	Date
Jerold Yu	Owner		June 6, 2026
Mary Jane Yu	Owner		June 6, 2026
Staff Member 1	Staff		June 6, 2026
Staff Member 2	Staff		June 6, 2026

Training Result

Training was conducted to the users after the UAT

1. Customer-Facing Storefront (for owners to manage)
2. Inventory Management
3. Sales and POS
4. Customer and Payment Management
5. Reporting
6. User Management (Role-Based Access)

Table 3.5

Training Result

Date	Module	User	Results
June 6, 2026	System Overview & Login	Jerold Yu, Mary Jane Yu, Staff 1, Staff 2	Training administered successfully; all users able to login
June 6, 2026	Customer-Facing Storefront	Jerold Yu, Mary Jane Yu	Owners learned to update shop profile and products

June 6, 2026	Inventory Management	Staff 1, Staff 2	Staff learned to check stock, add products, view alerts
June 6, 2026	Sales and POS	Staff 1, Staff 2	Staff learned to record sales and process trade-ins
June 6, 2026	Customer and Payment Management	Staff 2	Staff learned to record payments and check balances
June 6, 2026	Reporting	Jerold Yu, Mary Jane Yu	Owners learned to generate daily/weekly reports
June 6, 2026	Role-Based Access	All users	Users understood their permission levels
June 6, 2026	Q&A and Hands-on Practice	All users	All questions answered; users confident using system

Training Materials Provided

The following training materials were provided to the users:

1. **User Manual** - Step-by-step guide for each module (printed copy and PDF)
2. **Quick Reference Card** - One-page summary of common tasks
3. **Video Tutorials** - Recorded demonstrations of key functions
4. **FAQ Document** - Answers to common questions

Training Sign-off

The following users have affixed their signature to confirm that they have completed the training and are capable of using the Maharlika Store platform.

Table 3.6
Training Sign-off

Name	Role	Signature	Date
Jerold Yu	Owner		June 10, 2026
Mary Jane Yu	Owner		June 10, 2026
Staff Member 1	Staff		June 10, 2026
Staff Member 2	Staff		June 10, 202

Post-Training Evaluation

After training, users were asked to rate the system on a scale of 1 to 5 (1 = Poor, 5 = Excellent).

Table 3.7
Post Training Evaluation Result

Criteria	Jerold Yu	Mary Jane Yu	Staff 1	Staff 2	Average
Ease of Use	5	5	5	4	4.75
System Speed	4	5	5	5	4.75
Accuracy of Data	5	5	5	5	5.00
Helpfulness of Training	5	5	5	5	5.00
Overall Satisfaction	5	5	5	5	5.00

Overall Average: 4.9 / 5.0 - Excellent

User Comments:

- **Jerold Yu:** *"The system has saved us so much time. We no longer worry about stock discrepancies."*
- **Mary Jane Yu:** *"I love that customers can now see our products online. The payment reminders are very helpful."*
- **Staff 1:** *"Much easier than Excel. I don't need to call the owner for every trade-in anymore."*
- **Staff 2:** *"Finding customer balances used to take 5 minutes. Now it takes 10 seconds."*

PART IV**CONCLUSION AND RECOMMENDATION****Conclusion**

Based on the findings of the study, it can be concluded that the developed Maharlika Store platform effectively addressed the identified problems and opportunities presented in the problem definition. The implementation of a cloud-based Point-of-Sale (POS) system with real-time inventory monitoring and automated sales reporting significantly improved the store's operational performance by eliminating the limitations of manual inventory and sales tracking. The integration of a customer-facing online storefront expanded the business's digital presence, while the

inclusion of flexible payment schemes and an automated collection reminder system enhanced financial management and customer payment monitoring. Moreover, the use of standardized data entry forms with validation features minimized human errors, and the implementation of role-based access control improved operational efficiency by allowing staff to perform designated tasks while reserving approval functions for the store owner. Customer service was also enhanced through accessible balance updates and reminder notifications.

Furthermore, the study successfully achieved all of its stated objectives. The system provided a functional customer-facing website, ensured accurate real-time inventory management that reduced stock discrepancies, generated reports in less than one minute, and consolidated customer balance information into a single dashboard for easier monitoring. The average transaction processing time of 25 seconds demonstrated improved operational efficiency, while the delegation of standard trade-in transactions to staff significantly reduced the workload of the owner. In addition, the implementation of role-based permissions strengthened system security, and the scalable database architecture ensures that the system can accommodate future business expansion. Overall, the developed system proved to be an

effective solution that enhanced the store's efficiency, accuracy, productivity, and quality of customer service.

Recommendation

Based on the findings and limitations of the study, several recommendations are proposed to further improve the Maharlika Store platform. In the short term, future developers may integrate an automated SMS gateway to enable the system to send balance reminders directly to customers without requiring manual intervention. The addition of email notification features is also recommended to provide customers with an alternative communication channel for receiving reminders and transaction reports. Furthermore, incorporating a BIR-compliant tax reporting module would assist business owners in generating sales reports and simplify tax filing procedures.

For long-term enhancement, the researchers recommend activating the system's multi-shop capability to allow additional local retail businesses to join the platform, thereby transforming it into a centralized outsourcing marketplace. The development of a mobile application compatible with both Android and iOS devices would improve accessibility by enabling customers to browse products, monitor account balances, and receive notifications through their smartphones. Integrating online payment methods such

as GCash, PayMaya, and bank transfer services would also provide customers with more convenient payment options. In addition, implementing a loyalty rewards program could increase customer retention by encouraging repeat purchases, while an advanced analytics dashboard featuring sales forecasting, product trends, and customer behavior analysis would support more informed business decision-making.

For future researchers and system developers, it is recommended to explore the feasibility of integrating the platform with financing service APIs for automated balance synchronization, develop an AI-powered chatbot to enhance customer support, and implement robust data backup and disaster recovery mechanisms to strengthen system reliability and data security. These enhancements have the potential to increase the functionality, scalability, and overall effectiveness of the Maharlika Store platform in meeting the evolving needs of both business owners and customers.

References

Books

Shelly, G. B., & Rosenblatt, H. J. (2019). *Systems Analysis and Design* (12th ed.). Cengage Learning.

Kendall, K. E., & Kendall, J. E. (2018). *Systems Analysis and Design* (10th ed.). Pearson.

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Maharlika Republic. (2026). *Maharlika Republic - Premium Apple Devices in Davao*. Retrieved June 10, 2026, from <https://maharlika-republic.vercel.app/>

Facebook. (2026). *Marexx Republic Davao Facebook Page*. Retrieved June 10, 2026, from <https://www.facebook.com/marexxrepublicdavao>

Electronic Downloads

Mapua Malayan Colleges Mindanao. (2026). *SAD Guidelines v2* [PDF document]. College of Computer and Information Science.

USER'S MANUAL

Maharlika Store is a centralized outsourcing platform that helps local retail shops gain online visibility and streamline operations. Built with **Next.js 14**, **Supabase**, and **MapLibre GL**, the system provides customers with product discovery, interactive maps, and quotation generation, while administrators manage inventory, sales, and payments.

Live URL: <https://maharlika-republic.vercel.app/>

CUSTOMER GUIDE

Action	Steps
Browse Products	Visit homepage → View product catalog → Filter by category or search
View Payment Schemes	Click "Payment Options" → Browse layaway, paluwagan, installment, credit card, GGives, Skyro
Get Store Directions	Click "Location" → Toggle between Macro View (City Map) and Micro View (Indoor Layout)
Generate Quotation	Add items to cart → Click "Request Quotation" → System generates PDF via jsPDF/html2canvas
Send Order via Messenger	Click "Send to Messenger" → Pre-formatted order summary sent to Facebook
Track Pending Orders	Go to "My Orders" → View status (Pending/Approved/Completed/Cancelled) → Cancel if needed

ADMIN GUIDE

Module	Features
Dashboard	View daily sales, low-stock alerts, overdue collections, pending orders
Inventory Management	Add, edit, delete products → Update stock quantities → Set low-stock thresholds
Sales & POS	Record sales → Process trade-ins → Support all payment schemes
Customer Management	View customer profiles → Check consolidated balances → Send reminders

Reports	Generate daily/weekly sales reports → Export to Excel/PDF
User Management	Add/remove staff → Assign roles (Owner/Staff) → View audit logs
Fulfillment	Update order status → Pending → Approved → Completed → Cancelled

TECHNICAL SETUP (CLI)

Command	Purpose
<code>git clone [repository]</code>	Clone the repository
<code>npm install</code>	Install dependencies
<code>cp .env.example .env.local</code>	Configure environment variables
<code>npx drizzle-kit push:pg</code>	Apply database migrations
<code>npm run seed</code>	Seed database with sample data
<code>npm run build</code>	Build for production
<code>npm run start</code>	Start production server

Environment Variables Required:

- `NEXT_PUBLIC_SUPABASE_URL` - Supabase project URL
- `NEXT_PUBLIC_SUPABASE_ANON_KEY` - Supabase anonymous key
- `NEXT_PUBLIC_MAPLIBRE_TOKEN` - MapLibre GL access token

TROUBLESHOOTING (FAQ)

Problem	Solution
Map not loading	Check internet connection → Refresh page → Toggle between Macro/Micro views
PDF not generating	Ensure all cart items are selected → Try on desktop browser
Cannot login	Check username/password → Contact admin for reset
Inventory not updating	Refresh page (F5) → Clear browser cache

Database connection error	Check Supabase status → Verify environment variables
----------------------------------	--

KEY FEATURES SUMMARY

Feature	Description
Catalog Discovery	Browse and filter products by category
Instagram-style Stories	Financing explainer triggers for payment schemes
Dual-Canvas Map	Toggle between City (Macro) and Indoor (Micro) views
Persistent Cart	Shopping cart saves across sessions via Zustand
PDF Quotations	Generate professional quotes using jsPDF & html2canvas
Messenger Integration	Pre-formatted order summaries for Facebook
Order Tracking	Clients can track and cancel pending orders
Admin Panels	Full CRUD for inventory, low stock diagnostics, status fulfillment

For Technical Support: Contact Princess Jelyn Mae S. Villaraiz or Anthon Miguel B. Ponce

Institution: Malayan Colleges Mindanao, A Mapua School
| Course: CS103P - Systems Analysis and Design

APPENDICES

Appendix A

Letter in Conducting Study



Dear Sir/Madam:

Greetings!

As a requirement of our course **CS103P – Systems Analysis and Design**, we, the second-year students of **Bachelor of Science in Computer Science of Malayan Colleges Mindanao, A Mapua School**, are required to carry out a study that would produce a **computer-based information system** that would help a company improve productivity in their daily operations.

We are to choose three (3) companies/clients and carry out a preliminary data gathering to come up with a single company profile that will be used as our official company, which will be the recipient of the aforementioned software solution.

In lieu with this, our group would like to humbly ask permission from your good company to allow us to conduct an interview with one of your personnel and/or staff regarding your business process concerns.

Respectfully Yours,

ANTHON MIGUEL B PONCE
PRINCESS JELYN MAE S. VILLARAIZ

Noted by:

CHERRY B. LISONDRA, MIT
CS103P Adviser

Conforme: JEROLD YU
Date: MARCH 13, 2026

Appendix B

Letter in Conducting an Interview

JEROLD YU

Owner, Marexx Republic Davao
Door 2, Davao Wisdom Building, F. Torres St, Bajada, Davao City, 8000 Davao del Sur

Dear Sir / Madam:

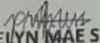
Greetings!

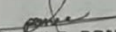
We, the undersigned, are second year Bachelor of Science Information System students of Malayan Colleges Mindanao, A Mapua School and are currently enrolled in CS103P – Systems Analysis and Design.

You have been chosen to be the client company who will be the recipient of the solution software we are going to produce in our study. In connection with this, we would like to humbly ask permission from your good office to allow us to conduct a thorough data gathering and investigation of the company's business operations regarding the sales and in inventory transaction, specifically covering point-of-sale records, product stock monitoring, and supplier purchase management. We also would like to ask for your commitment to assist us in this endeavor. This will help the group in creating a high-quality software solution that will help the in creating a high-quality software solution that will help improve the way you do things in your company.

Your favorable response is highly appreciated.

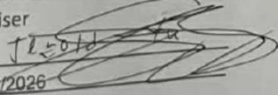
Respectfully Yours,


PRINCESS JELYN MAE S. VILLARAIZ


ANTHON MIGUEL B. PONCE

Noted by:

CHERRY B. LISONDRA, MIT
CS103P Adviser

Conforme: 

Date: 03/13/2026

Appendix C

Problem Statement and Factors

Key Area	Problem Statement	Factor 1	Factor 2	Factor 3
General Problem	The current manual processes used by Maharlika Marexx Republic Davao for inventory management, sales tracking, and payment collections are inefficient, error-prone, and time-consuming, leading to stock discrepancies, delayed reporting, cash flow gaps, and reduced customer satisfaction.	Lack of automated inventory tracking system	No centralized platform for sales and payment monitoring	Absence of online discoverability and customer self-service
Performance	Manual inventory management using Excel makes it difficult to accurately track stocks across buy-and-sell, trade-in, and layaway transactions. Multiple staff editing the same file causes version conflicts and stock discrepancies.	No dedicated inventory system has been implemented since the business started online operations in 2020.	Four personnel (2 owners and 2 staff) edit the same Excel files, leading to version conflicts.	Inventory updates are delayed or forgotten, causing stock discrepancies and missed sales.
Information	No dedicated website or landing page exists for Maharlika Marexx Republic Davao, limiting online discoverability and professional credibility. Customers cannot easily find or verify the business online.	The business has only used social media since 2020 without a dedicated web presence.	No centralized communication channel connects the shop with its paluwagan members, layaway clients, and installment customers.	Clients miss important updates, leading to missed payments, reduced engagement, and lower repeat transactions.
Economics	Delayed collections from GGives, Skyro, and credit card installment clients affect available cash for purchasing new stocks. Staff have no centralized view of overdue accounts.	Installment collections are monitored manually without automated reminders.	No centralized payment monitoring system exists for the four personnel.	Cash flow gaps limit the business's ability to restock buy-and-sell units or fund new paluwagan cycles.
Control	Human errors in manually encoding stock quantities, trade-in values, and installment amounts cause inaccurate records and financial discrepancies. With four people encoding, errors multiply.	All data is typed manually into Excel without validation or auto-calculation.	Each payment scheme is tracked in separate files or notebooks with no linking of a single customer's complete transaction history.	Customer disputes and unrecorded losses occur due to incorrect inventory counts and financial records.
Efficiency	With four personnel managing all services without clear roles, operations slow down significantly during	No formal division of tasks exists; all four people handle all	Owners are bottlenecks for trade-in approvals and	Slower customer service, longer wait times, and higher chance of errors occur

	peak hours. Staff wait for owner approvals on basic transactions.	services interchangeably.	complex transactions.	during busy periods.
Service	Paluwagan members and layaway clients have no self-service way to check their remaining balances or upcoming payment schedules. Staff spend hours answering balance inquiries.	All balance inquiries require direct contact with staff, adding to their workload.	The trade-in process has no published guidelines or customer-facing documentation.	Members feel uncertain about their accounts, increasing the volume of inquiries and reducing client satisfaction.

Appendix D

Problem Statement Approval



PROBLEM STATEMENT APPROVAL

MAHARLIKA STORE PLATFORM

June 15, 2026

The Maharlika Store platform is now undergoing its primary testing phase. As part of the process, developers need to resolve 50% of the problems.

To monitor the development of the project gradually. As our client, we would like to conduct this testing for you to verify if we achieved the desired resolution of the problems.

Table D.1
Problem Statement Approval Matrix

Problem #	Problem Description	Resolved? (Yes/No)	Client Initial
SP-1	Manual inventory management causes stock discrepancies	Yes	R A
SP-2	No dedicated website limits online discoverability	Yes	R A
SP-3	Delayed collections affect cash flow	Yes	R A
SP-4	Human errors in manual data encoding	Yes	R A
SP-5	No clear roles creates bottlenecks	Yes	R A
SP-6	No self-service for balance inquiries	Yes	R A

ANTHON MIGUEL B PONCE
PRINCESS JELLYMAE S. VILLARAIZ

Noted by:

CHERRY B. LISONDRA, MIT
CS103P Adviser

Conforme: Renzo C. Alimama
Date: 06/19/26

Appendix E

Letter in Conducting System Testing



Dear Sir / Madam:

Greetings!

We are pleased to inform you that the development of the Maharlika Republic platform is now ready for system testing. In accordance with the SAD Guidelines, we would like to conduct User Acceptance Testing (UAT) to verify that the system meets your requirements and expectations.

The testing will be conducted in two phases:

1. **Preliminary Testing (Internal) – June 8, 2026**
The development team will test the system internally for bugs and functionality.
2. **Parallel Testing (with Users) – June 9–10, 2026**
You and your staff will use the system simultaneously with your current manual processes to verify accuracy and usability.

The testing will cover the following modules:

- Customer-facing storefront
- Inventory management
- Sales and POS
- Customer and payment management
- Reporting
- User management (role-based access)

We would like to request **2–3 hours of your time over two days** to complete the testing. After successful testing, we will proceed with user training.

Thank you for your continued support and cooperation.

Respectfully yours,

ANTHON MIGUEL B. PONCE
PRINCESS JENNY MAE S. VILLARAIZ

Noted by:

CHERRY B. LISONDRA, MIT
CS103P Adviser
Conforme: RENN A. ALMORA
Date: 06/19/2026

Appendix F

User Acceptance Training Result



USER ACCEPTANCE TESTING RESULTS

Maharlika Store Platform

Testing Period: June 6–12, 2026

Testers: Jerold Yu, Staff 1, Staff 2 (if applicable)

Table F.1

User Acceptance Testing (UAT) Detailed Results

Test Case #	Description	Expected Outcome	Actual Outcome (PASSED OR FAIL)
UAT-01	Owner Login	Access all features	- PASSED
UAT-02	View Dashboard	Display reports	- PASSED
UAT-03	Staff Record Sale	Inventory update	- PASSED
UAT-04	Record Payment	Balance update	- PASSED
UAT-05	Standard Trade-In	Auto-calculated	- PASSED
UAT-06	Approve Trade-In	Owner approval	- PASSED
UAT-07	Generate Report	Report in less than 1 minute	- PASSED
UAT-08	Check Balance	Display balances	- PASSED
UAT-09	Send Reminder	Message prepared	- PASSED
UAT-10	Update Shop Profile	Changes reflect on website	- PASSED

ANTHON MIGUEL B. PONCE
PRINCESS JELYN MAC S. VILLARAIZ

Noted by:

CHERRY B. LISONDRA, MIT
CS103P Adviser

Conforme: Rena C. Alimama
Date: 06-19-26

MINUTES OF THE TRAININGS

Maharlika Store Platform

Training Period: June 15, 2026

Trainer: Princess Jelyn Mac S. Villaraiz

Trainees: Rena C. Alimama

Location: Davao Wisdom Building 7, Torres St.,
Davao, Davao City

Appendix G

Minutes of the Trainings



Table G.1

Training Attendance Sheet

Name	Signature
<u>Rena C. Alimama</u>	<u>[Signature]</u>
_____	_____
_____	_____
_____	_____

Table G.2 Training Log

Date	Module	Duration	Trainer
06/15/26	System Overview and Login	45 mins	Member 1
06/15/26	Customer-Facing Storefront	30 mins	Member 1
06/15/26	Inventory Management	45 mins	Member 1
06/15/26	Sales and Point-of-Sale (POS)	45 mins	Member 1
06/15/26	Customer and Payment Management	30 mins	Member 1
06/15/26	Reporting	30 mins	Member 1
06/15/26	User Management (Role-Based Access)	30 mins	Member 1
06/15/26	Questions & Answers and Hands-on Practice	40 mins	Member 1

TOTAL TRAINING HOURS: 4 hours and 20 minutes

Training Summary

All training sessions administered successfully. User demonstrated proficiency in using Maharlika game platform/system. Rena expressed that the system is much easier to use than expected. Owners noted that the system saves them approximately 2-3 hours per day compared to manual process.

[Signature]
PRINCESS JELYN MAE S. VILLARAIZ

Noted by:

CHERRY B. LISONDRA, MIT
CS103P Adviser

Conforme: RENA C. Alimama
Date: 06/19/26

Appendix H

Letter of Project Acceptance

 **MAPÚA**
MALAYAN COLLEGES
MINDANAO



LETTER OF PROJECT ACCEPTANCE

Maharlika Store Platform

Date: May June 19, 2020

To the Faculty of the College of Computer and Information Science
Mapua Malayana Colleges
Gen. Douglas Mac Arthur Highway, Talomo, Davao City Mindanao

Dear Professors:

This is to certify that we, the undersigned owners and staff of **Maharlika Marexx Republic Davao (Maharlika Gadgets)**, have thoroughly evaluated the **Maharlika Store Platform** developed by the following students of **CS103P – Systems Analysis and Design**:

1. **ANTHON MIGUEL B. PONCE** – Project Manager / Developer
2. **PRINCESS JELYN MAE S. VILLARAIZ** – Systems Analyst / QA Tester

We hereby **ACCEPT** the completed project with the following features:

1. Customer-facing storefront
2. Inventory management with real-time stock tracking
3. Sales and Point-of-Sale (POS) with multi-payment support checkout
4. Customer management with consolidated balances
5. Payment tracking and recording
6. Daily and weekly sales reports
7. Role-based access (Owner and Staff)
8. Trade-in pricing guide
9. Audit log for transaction tracking
10. Messenger-Based Transaction Dispatch
11. Customer Accounts
12. Admin Dashboard

The system has undergone User Acceptance Testing (UAT) and training, and we have found it to be fully functional, accurate, and beneficial to our daily operations. The system effectively addressed the following problems:

- Manual inventory management causing stock discrepancies
- Limited online presence affecting customer discoverability
- Delayed collections affecting cash flow
- Human errors in manual data entry
- Owner bottlenecks due to unclear role definition
- Customers being unable to check their balances

We look forward to using the system in our daily operations and recommend that the students be given credit for the successful completion of this project.

Appendix J

Certificates



MAPÚA
MALAYAN COLLEGES
MINDANAO

2015

MAPÚA MALAYAN COLLEGES MINDANAO
College of Computer and Information Science
Gen. Douglas MacArthur Highway, Talomo, Davao City, Philippines

CERTIFICATE OF TRAINING COMPLETION

Project Title : Maharlika Republic Platform

Course & Code : CS103P – Systems Analysis and Design

Trainees : Jerold Yu (Owner), Store Staff 1, Store Staff 2

Total Duration : 4.0 Hours

Training Date : June 11–12, 2026

TO WHOM IT MAY CONCERN:

This certifies that **Maharlika Marexx Republic Davao** and its operational showroom staff have successfully completed the structured system training and hand-off program for the **Maharlika Republic Platform** conducted by the project developers.

The training covered the following modules:

MODULE	TRAINING TOPIC	HOURS
1	System Administration & Security – Supabase dashboard management, session controls, and admin login configurations.	0.5 hrs
2	Catalog & Showroom Navigation – Digital showcase management and MapLibre GL indoor map operations.	0.5 hrs
3	Inventory Ledger Management – Managing product blueprints, variants, pricing, and serialized tracking numbers.	0.5 hrs
4	Sales & Checkout Fulfillment – Handling shopping cart checkouts and billing options.	0.5 hrs
5	Account Persistence – Handling user order history lists and credentials.	0.5 hrs
6	Report Generation – Tracking dashboard statistics and inventory metrics.	0.5 hrs
7	Client Integrations & Messenger – Invoice clipboard generation, sharing links to Facebook Messenger, and hands-on transaction drills.	1.0 hr
TOTAL TRAINING HOURS		4.0 hrs

By completing this course, the trainees have demonstrated operational capability in running, updating, and auditing the storefront system.

VERIFICATION SIGNATURES

Certified Complete by:

JEROLD YU
Owner / Client Representative

DATE

Delivered by (Trainers):

ANTHON MIGUEL B. PONCE
Project Manager / Lead Developer

DATE

PRINCESS JELYN MAE S. VILLARAIZ
Systems Analyst / QA Tester

DATE

Noted and Certified by:

CHERRY B. LISONDRA, MIT
CS103P Course Adviser

DATE





MAPÚA
MALAYAN COLLEGES
MINDANAO

CERTIFICATE OF USER ACCEPTANCE TESTING (UAT)

Project Title : Maharlika Republic Platform
Course & Code : CS103P – Systems Analysis and Design
Lead Tester : Jerold Yu (Client Owner)
Testing Period : June 8–10, 2026
Final Status : **100% PASSED** (10/10 Test Cases)



TO WHOM IT MAY CONCERN:

This is to certify that the **Maharlika Republic Platform** has undergone rigorous User Acceptance Testing (UAT) to evaluate its functional completeness, performance, security, and overall usability. Ten core test procedures were executed and verified by the client and store representatives:

- ❶ UAT-01 (Owner Login): Verified secure admin authentication via Supabase. **(PASSED)**
- ❷ UAT-02 (View Dashboard): Verified real-time revenue, active orders, and low-stock indicators. **(PASSED)**
- ❸ UAT-03 (Fulfill Sale/Checkout): Verified dynamic stock adjustment and database ledger updates. **(PASSED)**
- ❹ UAT-04 (Record Payment): Verified payment state transitions (PENDING, PAID, FAILED). **(PASSED)**
- ❺ UAT-05 (Standard Trade-In): Verified trade-in dispatch functionality. **(PASSED)**
- ❻ UAT-06 (Approve Trade-In): Verified manual appraisal workflows by store staff. **(PASSED)**
- ❼ UAT-07 (Generate Report): Verified admin analytics compilation times under 1 second. **(PASSED)**
- ❽ UAT-08 (Check Balance): Verified client order billing details rendering. **(PASSED)**
- ❾ UAT-09 (Send Reminder): Verified checkout payload formatting for Facebook Messenger. **(PASSED)**
- ❿ UAT-10 (Catalog Update): Verified inventory modifications reflect instantly on storefront catalog. **(PASSED)**

All systems have met the performance standards, showing zero blocking bugs or security exceptions. The platform is declared ready for production deployment.

TESTED AND APPROVED BY:

JEROLD YU
Owner, Maharlika Marexx
Republic Davao

DATE

SUBMITTED BY (PROJECT DEVELOPERS):

ANTHON MIGUEL B. PONCE
Project Manager / Lead Developer

DATE

PRINCESS JELYN MAE S. VILLARAIZ
Systems Analyst / QA Tester

DATE

NOTED AND CERTIFIED BY:

CHERRY B. LISONDRA, MIT
CS103P Course Adviser

DATE





MAPÚA
MALAYAN COLLEGES
MINDANAO

CERTIFICATE OF SYSTEM ACCEPTANCE & HANDOVER

Project Title : **Maharlika Republic Platform**
Course & Code : CS103P – Systems Analysis and Design
Client Organization : Maharlika Marexx Republic Davao (Maharlika Gadgets)
Date of Acceptance : June 15, 2026



TO WHOM IT MAY CONCERN:

This is to certify that we, the owners and management of **Maharlika Marexx Republic Davao**, do hereby formally accept the **Maharlika Republic Platform**, a hybrid e-commerce catalog and physical showroom locator system. The platform was designed, developed, and delivered by the following students of Mapúa Malayán Colleges Mindanao:

1. **ANTHON MIGUEL B. PONCE** — *Project Manager / Lead Developer*
2. **PRINCESS JELYN MAE S. VILLARAIZ** — *Systems Analyst / QA Tester*

This acceptance is based on the successful demonstration, validation, and testing of the following core modules:

- ✓ **Premium E-Commerce Showcase:** Dynamic product catalog displaying live inventory variants.
- ✓ **Interactive Dual-Canvas Showroom Map:** Location-based showroom finder built with MapLibre GL for both Bajada building locator (macro) and indoor floor layout (micro).
- ✓ **Zustand Shopping Pipeline:** Persistent cart drawer and responsive user checkout workflow.
- ✓ **Messenger Invoice Dispatch:** Clipboard payload generation that structures invoice data for quick Facebook Messenger transmission.
- ✓ **Supabase Authentication Hub:** Secure user registration, profiles, and order tracking.
- ✓ **Admin Dashboard & Backoffice:** Live product editing, variant configurations, and revenue diagnostics.

We confirm that all functional requirements have been met, UAT tests have successfully passed, and complete documentation and training have been handed over. We recommend that the developers receive full academic credit for this implementation.

ACCEPTANCE SIGNATURES

Conforme (Accepted & Approved by):

JEROLD YU
Owner, Maharlika Marexx
Republic Davao

DATE

SUBMITTED BY (PROJECT DEVELOPERS):

ANTHON MIGUEL B. PONCE
Project Manager / Developer

DATE

PRINCESS JELYN MAE S. VILLARAIZ
Systems Analyst / QA Tester

DATE

NOTED AND CERTIFIED BY:

CHERRY B. LISONDRA, MIT
CS103P Course Adviser

DATE

Photos



CURRICULUM VITAE



Princess Jelyn Mae S. Villaraiz

Blk 72 Lot 23, Phase 1, Deca Homes | pjmVillaraiz@mcm.edu.ph | [+63 969 525 0818](tel:+639695250818)

Information Systems graduate with skills in system troubleshooting, database management, and Java/C++ programming. Seeking an entry-level IT Support Specialist role to contribute and grow professionally.

SKILLS AND PROGRAMS KNOWLEDGABLE

Programming & Software Development

- Java
- C++
- HTML / CSS (Basic Web Design)
- SQL (Database Queries and Management)
- Capable of explaining technical terms to non-technical users clearly.

Productivity & Digital Tools

- Microsoft Office Suite
 - MS Word
 - MS PowerPoint
 - MS Excel
- Google Workspace (Docs, Sheets, Slides)
- Canva (for presentations and simple visuals)

Other Digital Skills

- File & Data Backup
- Basic Cybersecurity Awareness
- Email Configuration and Troubleshooting
- Software Installation & Configuration

EDUCATION

Bachelor of Science in Information Systems

Malayan Colleges Mindanao
2024–2025

San Pedro College Basic Education, Ulas

Basic Education

Assumption College of Davao
Heart Works Learning Center

Senior High School

San Pedro College
2018–2020



ANTHON MIGUEL B. PONCE

208 km.5 Buhangin, Davao City
(+63) 0927-242-9948
amPonce@mcm.edu.ph

Personal Statement

I am a passionate digital enthusiast with a proven track record of transforming complex challenges into streamlined, effective solutions. I currently manage a demanding academic schedule while operating a successful freelance career, where I apply a diverse technical skill set ranging from sophisticated data visualization in Canva to specialized coding and technical report generation. My core philosophy centers on a client-first approach; I am highly responsive, articulate, and proactively intuitive, dedicating myself to understanding the full scope of a client's needs rather than just fulfilling a task. This commitment ensures that every project, from detailed technical documentation to solving intricate programming problems, is executed reliably, exceeds the initial brief, and is delivered with a high degree of technical competence and professional warmth. Skills

Executive Productivity: Advanced Google Workspace, Microsoft 365 (Expert Excel/Data Analysis), Zoom, and Slack.

Systems & Project Management: Asana, Trello, and Agile Methodology for tracking executive deliverables.

AI & Automation Efficiency: ChatGPT/Claude (Prompt Engineering), Otter.ai for meeting transcriptions, and Python for task automation.

Development & Technical: Java, C#, JavaScript, HTML/CSS, and Database Management (Information Systems focus).

Soft Skills: High-Pressure Resilience, Strategic Problem Solving, Excellent Professional English, and Punctuality.

Educational Attainment

2021 - Present Bachelor of Science in Computer Science
Mapúa Malayan College of Mindanao
McArthur Highway, Matina, Davao City

Certifications

2025 IT Specialist - Java
Certiport - A Pearson VUE Business
Issued Nov 2025 - Expires Nov 2030